KARTIK VEGAD
INTERNSHIP - SAHANA SYSTEM PVT. LTD.

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COMPUTER VISION

WHAT IS COMPUTER VISION?

Computer Vision is a field of Artificial Intelligence that enables computers to analyse and interpret visual information from images, videos and other visual inputs.

Humans naturally understand visual scenes through the human visual system. Computers however receive images only as numerical pixel data. Computer vision algorithms process these pixel values to extract meaningful patterns and information.

The goal of computer vision is to allow machines to understand visual data and make intelligent decisions based on it.

This capability allows machines to recognise objects understand scenes track motion and analyse visual patterns.

WHY COMPUTER VISION?

Visual data is one of the largest sources of information in the modern digital world. With the widespread use of cameras smartphones drones and surveillance systems enormous amounts of visual data are generated every day.

Computer vision enables machines to automatically analyse this visual data which helps in

- Automating visual inspection tasks
- Improving decision making systems
- Enhancing human computer interaction
- Supporting intelligent systems such as robots and autonomous vehicles

Because of these capabilities computer vision has become one of the most important research areas in modern artificial intelligence.

HOW CV WORKS?

01 Image Acquisition

A camera or sensor captures an image or video frame.

02 Image Processing

Basic image processing techniques manipulate pixel values to enhance image quality. Examples include filtering edge detection and smoothing.

03 Feature Extraction

Important structures such as edges textures corners or shapes are identified.

04 Model Learning

Machine learning or deep learning models learn patterns from training data. Modern systems widely use Convolutional Neural Network architectures for visual recognition tasks.

05 Prediction

The trained model interprets new images and produces outputs such as object labels or bounding boxes.

KEY CONCEPTS

Feature Extraction
Identifying meaningful patterns such as edges textures or shapes within an image.

Object Detection
Locating and classifying objects in an image. Modern models include YOLO and R-CNN.

Motion Analysis
Understanding movement in video data including object tracking and activity recognition.

Image Classification
Assigning an image to a predefined category using models such as CNNs.

Image Segmentation
Dividing an image into meaningful regions to identify objects and boundaries.

Image Processing
Manipulating image pixel values through operations such as filtering enhancement and segmentation.

3D Vision
Estimating depth and reconstructing three dimensional information from two dimensional images.

PROS

- Automation of visual analysis tasks
- Ability to process large amounts of image data quickly
- Improved accuracy in pattern recognition using deep learning
- Real time analysis of video streams
- Wide applicability across industries

CONS

- Large labelled datasets are often required for training models
- High computational resources are needed for training deep learning models
- Performance may be affected by lighting changes or occlusions
- Models may exhibit bias if training data is not diverse

REAL WORLD APPLICATIONS

Autonomous Vehicles

Detect road signs vehicles pedestrians and lanes.

Healthcare

Analyse medical scans for disease detection.

Retail

Checkout systems & inventory monitoring.

Security

Face recognition and surveillance systems.

Manufacturing

Quality inspection and defect detection.

Agriculture

Detect crop diseases and monitor plant health.

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THANK YOU